THE STRUCTURE MORPHISM OF A RELATIVE PROJ IS SEPARATED

THE STACKS PROJECT AUTHORS

Allowed standard prerequisites (editorial).

- Separatedness of Proj. lemma-proj-separated (carded this batch). Proj(A) is separated.
- Separatedness is local on the base. schemes-section-separation-axioms (cross-chapter, named). Separatedness of pi may be checked locally on S.
- Composition after a separated morphism. schemes-lemma-compose-after-separated (cross-chapter, named). Proj(A) to U separated composed with the separated U to S.

Source reference. Original lemma 01MC: sources/stacks-constructions/pinned-original-source.tex, source lines 1532–1532. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01KV: focused-seeds/geometry/S9-cons-relative-proj-separated/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01KH: focused-seeds/geometry/S9-cons-relative-proj-separated/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme and $\mathcal{A}$ be a quasi-coherent sheaf of graded $\mathcal{O}_S$-algebras. The morphism $\pi : \underline{Proj}_S(\mathcal{A}) \to S$ is separated.*

Proof. To prove a morphism is separated we may work locally on the base, see Schemes, Section 01KH. By construction $\underline{\mathrm{Proj}}_S(\mathcal{A})$ is over any affine $U \subset S$ isomorphic to $\mathrm{Proj}(A)$ with $A = \mathcal{A}(U)$. By Lemma 01MC we see that $\mathrm{Proj}(A)$ is separated. Hence $\mathrm{Proj}(A) \to U$ is separated (see Schemes, Lemma 01KV) as desired. □

Date: stacks-project:constructions.tex:local-snapshot-20261012; GNU FDL 1.2 or later.